THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET B

Chapter 13: **Our Home Earth, a Unique Life-Sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The region around a star where water can stay liquid is the: 1

(a) ozone layer **(b)** Kuiper belt **(c)** habitable zone **(d)** asteroid belt

2. If gravity were too strong, creatures would be: 1

(a) crushed by their own weight **(b)** floating
(c) taller **(d)** lighter

3. The greenhouse effect on Earth: 1

(a) removes the atmosphere **(b)** keeps Earth warm by trapping heat
(c) makes Earth colder **(d)** stops sunlight

4. The geosphere consists of the: 1

(a) living things **(b)** air
(c) water **(d)** solid part of the Earth

5. In asexual reproduction: 1

(a) gametes fuse
(b) a single parent produces offspring identical to itself
(c) two parents are always needed
(d) new features appear always

6. Transfer of pollen from anther to stigma is: 1

(a) germination **(b)** budding **(c)** pollination **(d)** fertilisation

7. In fish and frogs, fertilisation takes place: 1

(a) inside the body **(b)** in soil **(c)** in water **(d)** in air

8. Seeds can spread far when they are: 1

(a) left to rot in the plant **(b)** eaten only by humans
(c) kept in the plant **(d)** eaten by birds and dropped

9. A small rise in global temperature can: 1

(a) remove gravity **(b)** increase ozone
(c) disturb climate patterns **(d)** stop all seasons forever

10. Soil, rocks and minerals are part of the: 1

(a) hydrosphere **(b)** geosphere **(c)** biosphere **(d)** atmosphere

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. How does Earth's magnetic field help life? 2

12. Define atmosphere, hydrosphere, geosphere and biosphere in one line each (any two). 2

13. What is the greenhouse effect? 2

14. Name two greenhouse gases. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain asexual reproduction in plants with two examples. 3

16. What is regeneration? Give an example. 3

17. Explain pollination and fertilisation in flowering plants. 3

OR

Describe seed dispersal with an example. 3

18. Explain how sexual reproduction in animals differs in fish and mammals. 3

OR

Describe the development of a bird from fertilisation to hatching. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain sexual reproduction in flowering plants from pollination to seed formation. 5

OR

Compare reproduction in fish, frogs, birds and mammals on fertilisation and embryo development. 5

20. Discuss the triple planetary crisis and give actions to address it. 5

OR

Explain the role of the atmosphere in sustaining life: oxygen, ozone and greenhouse effect. 5

END OF QUESTION PAPER — Total: 40 Marks

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